

Precision Digital Agriculture Hackathon 2026 – Corn Breeding Track

0. Track at a Glance

Track name: Corn Breeding Track Focus: Applying genomic prediction and data integration to optimize breeding decisions under resource constraints in commercial corn development programs.

Themes: Multi-omics data integration; genomic prediction; genotype-by-environment interaction; breeding value estimation; resource allocation; commercial pipeline optimization.

Recommended team roles:

- Domain lead - plant breeding context, genetic principles, commercial feasibility
- Data/ML lead - modeling, cross-validation, uncertainty quantification, genomic prediction
- Software lead - repository management, data integration pipelines, reproducible analysis
- Communication lead - breeding impact interpretation, stakeholder recommendations, presentation

1. Background and Why This Track Matters

Commercial corn breeding operates under significant resource constraints while serving global food security needs. Breeding companies must evaluate hundreds of thousands of genetic combinations annually across diverse environments, but field testing capacity is limited and expensive. A single season of poor allocation decisions can cost millions in lost genetic gain and delayed variety releases.

Modern breeding programs integrate phenotypic evaluation, environmental characterization, and genomic data to predict line performance and optimize resource allocation. However, data integration challenges, genotype-by-environment interactions, and prediction uncertainty complicate these decisions. The industry needs robust prediction systems that can guide breeding decisions under real commercial constraints.

Your prototype must address the fundamental challenge of translating complex multi-omics datasets into actionable breeding recommendations. Success means demonstrating an

end-to-end workflow from raw breeding data to specific line advancement decisions, with transparent uncertainty quantification and clear documentation of model limitations and commercial applicability.

2. Global Prompt Overview

Your mission: Build a practical prediction system that integrates phenotypic, environmental, and genomic data to guide breeding line advancement decisions under resource constraints, using historical data from a commercial corn breeding program (2001-2007) to predict 2008 performance.

Scenario Context: Late Friday in January 2008, budget cuts have reduced available field plots for the upcoming season. The breeding team needs data-driven recommendations for which maize lines to advance from the 115 Relative Maturity pipeline.

Deliverables Required:

1. Problem statement (breeding decision context, stakeholder impact, commercial significance)
2. Solution overview (data integration workflow + prediction approach)
3. Technical approach (baseline models + your methodology; preprocessing; model validation)
4. Results (prediction accuracy; line ranking; uncertainty quantification)
5. Run instructions (clear "judge mode" with sample data subset)
6. Commercial recommendations (specific lines to advance, resource allocation strategy)
7. Constraints and limitations (model failures, data gaps, next development steps)

Data → Decision Framing: Multi-omics integration → Genomic prediction → Performance forecasting → Line selection → Commercial advancement

3. Dataset Overview

Commercial Context: This dataset represents the initial screening stage within the 115 Relative Maturity (RM) Pipeline - a commercial corn breeding program's general germplasm evaluation phase focused on General Combining Ability (GCA) assessment.

Breeding Program Structure: - Inbred development within clusters (C1×C1, C2×C2) - Testcross evaluation using opposite cluster testers - Multi-location, multi-year performance assessment (2001-2007) - Commercial scale: ~499 populations (C1), ~500 populations (C2)

Data Components

Phenotypic Data (2001-2007)

- C1PhenotypeData_V2.csv (141 MB): Phenotypic measurements for Cluster 1 individuals
- C2PhenotypeData_V2.csv (141 MB): Phenotypic measurements for Cluster 2 individuals

Key Traits: - YLD_BE: Yield (bushels per acre) - Primary target variable - MST: Moisture content at harvest - PHT: Plant height - EHT: Ear height - TWT: Test weight - RTLP/STLP: Root and stalk lodging percentages

Environmental Data

- environmental_features.csv (217 KB): Environmental conditions by year and location

Environmental Variables: - Climate: Temperature (TMAX, TMIN, TAVG), Precipitation (PRCP) - Soil: Texture (clay, silt, sand percentages), pH, nitrogen content - Geographic: Location codes, coordinates

Genomic Data

- ImputedPopulationsC1/: 499 populations with SNP marker data for Cluster 1
- ImputedC2Populations.zip: 500 populations with SNP marker data for Cluster 2

Data Structure: - Each population file contains progeny lines (000000000001, 000000000002, etc.) - First two rows in each file are parent lines (PID identifiers) - Thousands of SNP markers per individual - Genotype calls: 1, 0, -1 (different allelic states)

ID Matching System

LINEUNIQUEID Format: C{Cluster}.{Population}.{Line} - Example: C1.1.2 = Cluster 1, Population 1, Line 2 - Maps to genomic file: C1.1_Imputed.csv, individual "000000000002"

Linking Keys

1. Phenotype ↔ Environment: YEAR + LOC
2. Phenotype ↔ Genomic: Extract line number from LINE_UNIQUE_ID, convert genomic IDs to integers
3. Geographic: LONGITUDE + LATITUDE

4. Challenge Prompts

Choose one primary challenge below. You may combine approaches or develop alternative strategies, as long as your solution addresses the core resource allocation decision and provides actionable breeding recommendations.

Challenge 1: Broad-Acre Performance Prediction

Prompt: How can you predict general performance across environments to identify lines with stable, broadly-adapted performance?

Approach Examples: - Multi-environment yield prediction using genomic and environmental data - Stability analysis combining genotypic and phenotypic information - Risk-adjusted performance metrics that account for environmental variation

Prototype Output: Ranked line list with predicted performance and stability metrics, resource allocation recommendations.

Challenge 2: Environment-Specific Performance Prediction

Prompt: How can you predict location-specific performance to enable precision breeding recommendations based on local adaptation?

Approach Examples: - Genotype-by-environment interaction models - Environment-specific genomic prediction - Location-tailored breeding value estimation

Prototype Output: Environment-specific performance forecasts, targeted line recommendations by location.

Challenge 3: Multi-Trait Optimization Under Constraints

Prompt: How can you optimize breeding decisions when multiple traits matter and trade-offs exist between performance characteristics?

Approach Examples: - Multi-trait selection indices - Pareto frontier analysis for trait trade-offs - Economic weight integration for commercial value

Prototype Output: Multi-objective optimization results, trade-off analysis, breeding strategy recommendations.

5. Constraints and Considerations

Data and Privacy:

- Use only the provided breeding program datasets
- All genomic and phenotypic data are de-identified research datasets
- Sample data files provided for initial development and testing

Commercial Viability and Workflow Fit:

- Define your intended users (breeding program managers, population development teams, field testing coordinators)
- Make predictions actionable for 2008 planting season decisions
- Consider computational requirements for operational deployment

Validation, Uncertainty, and Prediction Reliability:

- Include baseline prediction models (phenotypic BLUP, environmental means)
- Implement cross-validation appropriate for breeding program structure
- Quantify prediction uncertainty and confidence intervals
- Address missing data challenges (incomplete genomic coverage, environmental gaps)
- Define failure modes (insufficient training data, novel environments, population structure shifts)

Technical Constraints:

- Large file sizes require efficient data handling strategies
- Population structure and family relationships affect model assumptions
- Genotype-by-environment interactions complicate broad predictions

- Historical data may not fully represent future conditions

6. Submission Package and Reproducibility Requirements

Your submission must include a README with:

- Problem statement and breeding decision context
- Solution overview and technical approach
- Model validation results and uncertainty analysis
- Commercial breeding recommendations

Additional Constraints:

- Do not upload raw large datasets; use provided sample files for demonstrations
- Provide clear installation and execution instructions

Judge-Mode Guidance for This Track:

- Include synthetic data generation if needed to showcase methodology
- Provide end-to-end pipeline that runs without requiring full 141MB datasets
- Document scaling approach for full dataset deployment

7. Resources and Technical References

Provided Sample Data Files:

- sample_C1_phenotype_100rows.csv: Subset of C1 phenotype data
- sample_environmental_20rows.csv: Environmental data sample
- sample_C1.1_100rows.csv: Genomic data sample for population C1.1

Technical Documentation:

- CORNBREEDINGDATA_GUIDE.md: Complete technical data documentation
- Hackathon Scenario And Help.docx: Breeding concepts and context
- Data Dictionary.docx: Variable definitions and protocols

■■ Getting Started

Recommended Approach

1. Start Small: Use sample files to understand data structure
2. Explore Relationships: Examine trait correlations, environmental patterns
3. Build Baseline Models: Simple models using key variables
4. Integrate Genomics: Add marker data for improved predictions
5. Validate Strategy: Cross-validation with historical data
6. Scale Solution: Apply to full dataset

■ Evaluation Criteria

Technical Excellence

- Model accuracy and robustness
- Appropriate use of available data types
- Statistical rigor and validation

Commercial Viability

- Practical implementation considerations
- Resource efficiency
- Risk management approach

Innovation

- Creative use of multi-omics data
- Novel analytical approaches
- Scalability of solution

Communication

- Clear presentation of methods and results
- Business impact justification
- Actionable recommendations